

Rishi More

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Summary

ML systems engineer specializing in efficient and reliable LLM inference. **ICML 2026 co-first author** with hands-on experience in CUDA/Triton kernels, Nsight profiling, INT4 quantization, vLLM serving, and distribution-free risk control.

Experience

JHU Data Science and AI Institute (DSAI)

May 2025 – June 2026

Graduate Research Assistant, LLM Inference Optimization

Baltimore, Maryland

- Co-first authored *Conformal Thinking (ICML 2026 main track)*, introducing finite-sample calibrated early stopping for reasoning LLMs while controlling a target error rate.
- Built a vLLM-based calibration and evaluation pipeline across **4 reasoning models, 4 benchmarks, and 2.7K+ problems**, maintaining target error rates under dataset and response-length shift with finite-sample UCB calibration.
- Designed dual-threshold and ensembled halt policies that reduced reasoning tokens by **12% at matched accuracy**, with reusable tooling for calibration, evaluation, and ablation studies.

JHU Center for Language and Speech Processing (CLSP)

January 2025 – December 2025

Graduate Researcher, Retrieval Systems

Baltimore, Maryland

- Built temporal RAG over **10K+ timestamped documents** using revision-aware chunk tracking, time-decay scoring, and metadata-scoped indexing to prioritize current evidence.
- Improved retrieval precision by **23%** over dense-only retrieval and reduced stale or hallucination-prone contexts by **31%** through recency weighting and reference-free consistency scoring.

Mastek

November 2022 – July 2023

Machine Learning Engineer Intern

Mumbai, India

- Deployed an on-premises edge voice-authentication system with **<100 ms p99 latency** and **96.6% accuracy**, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment.
- Built a TensorFlow pipeline over **1M+ voice samples**; spectral-gating denoising, augmentation, and robust feature extraction improved authentication reliability by **20%** in high-noise conditions.

Projects

Conformal Serve: Risk-Controlled Adaptive LLM Serving | vLLM, FastAPI, Docker Compose, Prometheus

- Built an engine-agnostic adaptive-stopping core shared by deterministic mock-engine CI and vLLM's AsyncLLMEngine; achieved **100% policy branch coverage** with `mypy --strict` and exposed client-set risk α through an OpenAI-compatible API.
- On Qwen2.5-1.5B-Instruct and an 8 GB RTX 4070, reduced reasoning tokens **64.7% at $\alpha = 0.10$** and **71.4% at $\alpha = 0.30$** while maintaining realized error $\leq \alpha$ on small held-out arithmetic sets.
- Calibrated per- α thresholds offline by distribution-free risk control into immutable, content-addressed artifacts; shipped Docker Compose with Prometheus/Grafana and a one-command benchmark reporting **95% Clopper-Pearson error bounds**.

INT4 KV-Cache Quantization & Fused Triton Attention for Long-Context Decode | Triton, CUDA, PyTorch

- Built a correctness-gated fused Triton INT4 dequantization-attention kernel, cutting 32K-context attention latency **13.7 $\times$ (7.54 to 0.55 ms/layer)** versus PyTorch FP16 GQA SDPA while maintaining **0.999 cosine similarity** to the dequantized oracle.
- Reduced KV storage from **28.7 KB to 8.3 KB per token (3.47 $\times$)**, raising estimated 8GB KV capacity from **$\sim 154\text{K}$ to $\sim 580\text{K}$ tokens**; retained **100% passkey retrieval through 32K** with only **+1.0% WikiText perplexity**.
- Diagnosed per-token-K failure as outlier-channel collapse (attention cosine **0.08 vs. 0.98**) and built an RCPS controller certifying **$\sim 50\%$ cache eviction at 19% realized loss under a 30% risk budget**.

Decode Roofline: Kernel Profiling & Fused INT4 Dequant-GEMV | CUDA C++, Nsight, vLLM

- Profiled Qwen2.5-1.5B batch-1 decode with Nsight; attributed **$\sim 81\%$ of GPU-kernel time** to weight GEMVs and measured dominant MLP projections at **$\sim 247\text{ GB/s}$** against a **$\sim 250\text{ GB/s}$** DRAM ceiling.
- Built a correctness-gated fused CUDA INT4 group-dequantization + GEMV kernel reaching **$\sim 223\text{ GB/s}$** ; accelerated batch-1 MLP projections by **2.38 $\times$** versus FP16 GEMM and mapped the crossover to batch 2 as weight reuse made GEMM faster.

Education

Johns Hopkins University

M.S.E. in Computer Science

May 2026

GPA: 3.9/4.0

University of Mumbai

B.Tech. in Computer Science, Honors in AI/ML

July 2024

CGPA: 9.85/10

Technical Skills

Languages: Python, C++, CUDA C++, C, SQL, Bash

ML/LLM Systems: PyTorch, Triton, vLLM, Hugging Face Transformers, TensorFlow, LoRA, DPO, RAG

Performance & Infrastructure: Nsight Compute/Systems, Linux, Git, Docker, GitHub Actions, AWS

Methods: Quantization, kernel fusion, roofline analysis, LLM evaluation, conformal risk control, calibration